NUFYP Mathematics
Spring Semester

Learning Block 2

2.2 Probability 1

NAZARBAYEV UNIVERSITY

Learning outcomes

2.2.1 State the definitions of outcomes, experiment, event, sample space and the probability of an event

2.2.2 Find the probability of an event using the definition

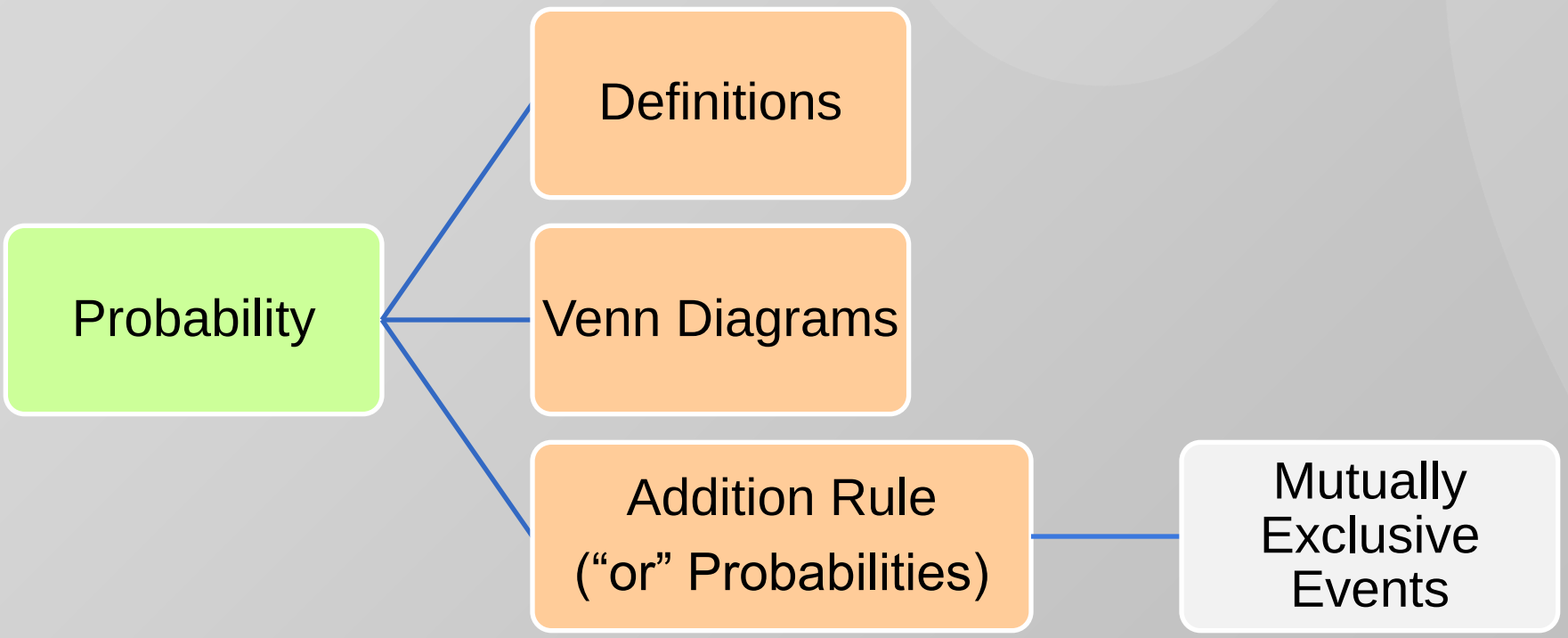
2.2.3 Interpret the regions of a Venn diagram and use Venn diagrams to represent events, and solve problems

2.2.4 Use the Addition Rule $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ to solve “or” probability problems

2.2.5 Describe the meaning of mutually exclusive events

2.2.6 Apply the Addition Rule to mutually exclusive events

Lecture outline



Introduction

Life is full of uncertainties!

What are the chances of rain tomorrow?

What are the chances of winning a lottery?

Probability indicates the likelihood of an event and it helps you make decisions.

Some real-life examples using probability are:

1. Weather forecasting
2. Sports strategies
3. Insurance plans
4. Games

What is the “probability”?

First, we need to learn the vocabulary.

- An **experiment** (or **trial**) is a repeatable process that has a set of possible **outcomes**.
- An **event** is a collection of one or more outcomes.
- The **sample space** is the set of all possible outcomes of an experiment. We use S to denote the sample space.

Example 1

Suppose you roll a six-sided die.

Rolling a die is an experiment.

Possible outcomes are 1,2,3,4,5,6.

"Landing on a 6" is an event: $A = \{6\}$

"Landing on an even number" is an event:

$$B = \{2,4,6\}$$

"Landing on a number less than 5" is an event:

$$C = \{1,2,3,4\}$$

The sample space is $S = \{1,2,3,4,5,6\}$

What is the "probability"?

Let A be an event. *Assuming all outcomes are equally likely*, the probability of A is

$$P(A) = \frac{n(A)}{n(S)} \quad , \quad \text{where}$$

$n(A)$ is the number of outcomes in the event A, and $n(S)$ is the number of outcomes in the sample space S.

<u>Note</u>: $P(S) = \frac{n(S)}{n(S)} = 1 = 100\%$

Example 2

- Flipping an *unbiased* coin is an experiment.

S = { H , T } ; H and T are equally likely.

Let A = { H }. Then, $P(A) = \frac{n(A)}{n(S)} = \frac{1}{2} = 50\%$.

- Flipping a *biased* coin is also an experiment.

Suppose a coin is <u>weighted</u> in a way that *heads* occurs more frequently than *tails*.

S = { H, T } still, but H and T are not equally likely.

Here, we can't apply the formula $P(A) = \frac{n(A)}{n(S)}$, and $P(A) \neq \frac{1}{2}$.

Note: Unless mentioned otherwise, we will assume that coins or dice are unbiased.

Example 3

When throwing a six-sided die, what is the probability of getting an even number?

Solution

$A = \{2,4,6\}$ and $S = \{1,2,3,4,5,6\}$.

Therefore, the probability is $\frac{n(A)}{n(S)} = \frac{3}{6} = \frac{1}{2} = 50\%$.

What is the probability of getting a 7?

- An **impossible** event has a probability of $\mathbf{0} = 0\%$.
- A **certain** event has a probability of $\mathbf{1} = 100\%$.
- So, all events have probabilities between 0 and 1, inclusive. That is, $0 \leq P(A) \leq 1$.
- Probabilities can be written as fractions, decimals or percentages.

Example 4

Two ***fair*** spinners each have four faces numbered 1 to 4. The two spinners are spun together and the sum of the numbers indicated on the two spinners is recorded.

Find the probability of the spinners indicating a sum of

a) exactly 5,

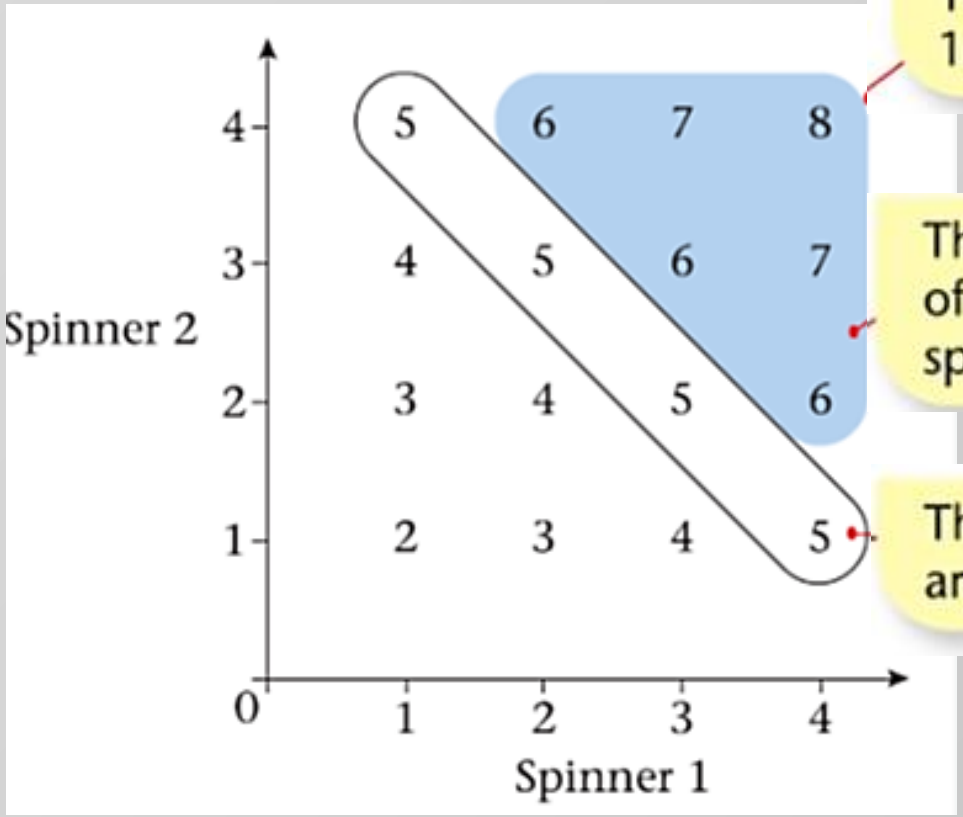
b) more than 5.

+

= ?

What does the word "fair" imply?

Solution



The sample space is represented by the 16 points on the diagram.

There are $4 \times 4 = 16$ points. Each of these points is equally likely as the spinners are fair.

There are 4 outcomes for this part. They are $1 + 4$, $2 + 3$, $3 + 2$ or $4 + 1$.

a) $P(\text{Exactly } 5) = \frac{4}{16} = \frac{1}{4}$

b) $P(\text{more than } 5) = \frac{6}{16} = \frac{3}{8}$

Example 5

What is the probability that an arrangement (chosen at random) of the letters from the word “WATER” starts with a vowel?

Solution

The total number of arrangements (outcomes) is 5! = 120.

The number of arrangements starting with vowel is:

2 (either A or E) x 4! = 48.

Hence, $\mathrm{P} = \frac{48}{120} = \frac{2}{5}$.

Your turn!

What is the probability that the repeated letters (i.e., indiscernible letters) in the word “blueberry” are placed next to each other?

Solution

There are two b’s, two e’s, and two r’s.

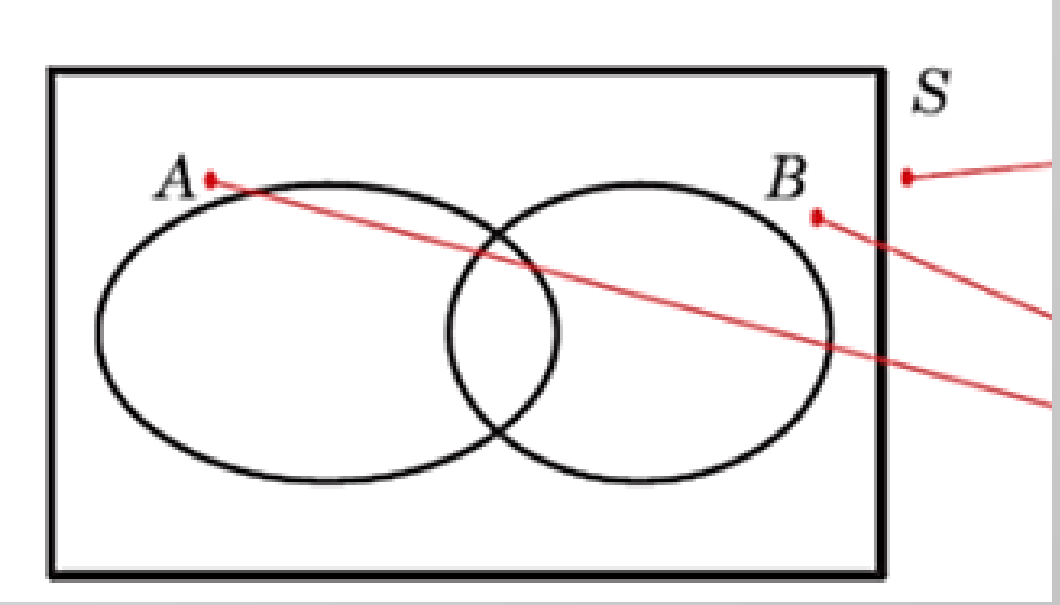
The total number of (discernible) arrangements is $\frac{9!}{2!2!2!}$.

Treating the same letters as one item, we have 6 items: 6!

The probability is $\mathrm{P} = \frac{6!}{\frac{9!}{2!2!2!}} = \frac{2!2!2!}{9\times8\times7} = \frac{1}{63}$.

Venn diagram

A Venn diagram is a drawing where the outermost box represents the sample space, S, and the circles inside represent events.



S represents the sample space. $P(S) = 1$

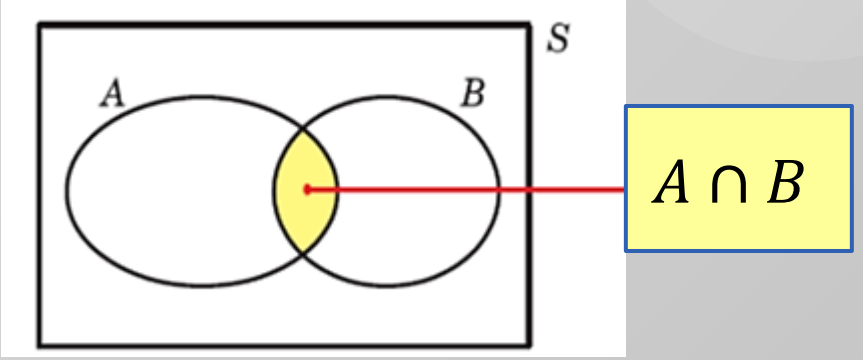
A and B represent events.

Venn diagrams provide an excellent visualization for probabilities.

Set notation

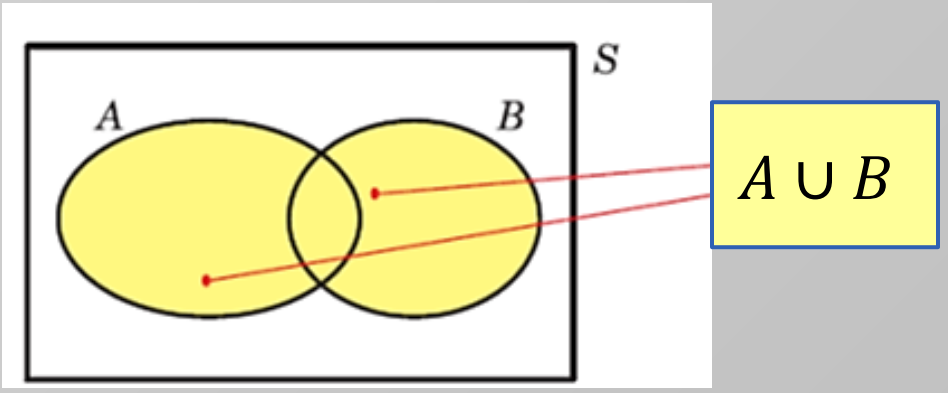
<u>Intersection</u>

A *and* B both occur: denoted by A ∩ B



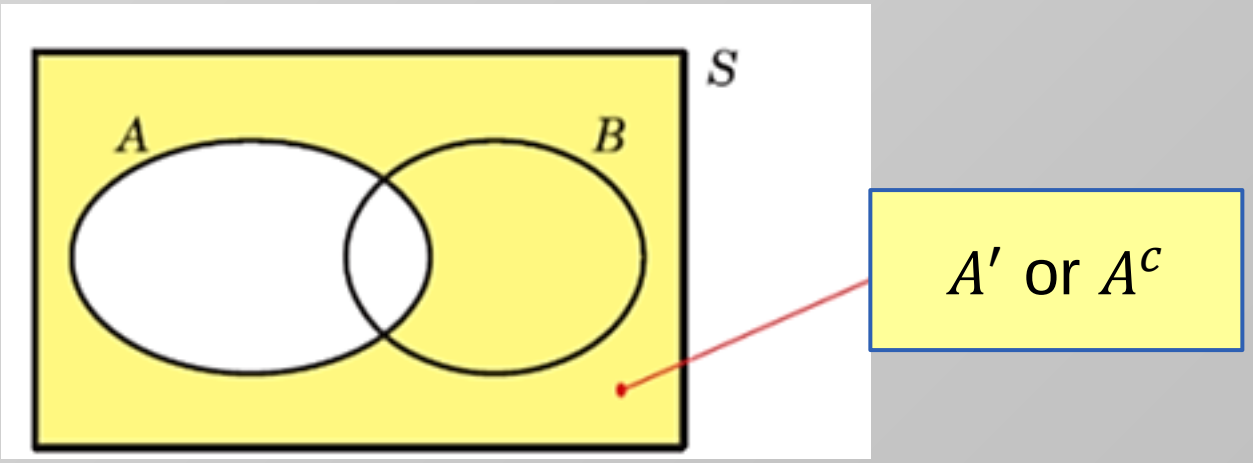
<u>Union</u>:

A *or* B *or* both occurs: denoted by A ∪ B



Set notation

<u>Complement</u> Any outcome *not in A* occurs:
sometimes denoted by A’
sometimes denoted by A^c



P(A’) = 1 – P(A)

Example 6

A survey of 79 families found that 22 own cats, 27 own dogs, and 7 own both cats and dogs.

a) Draw a Venn diagram to represent this information.

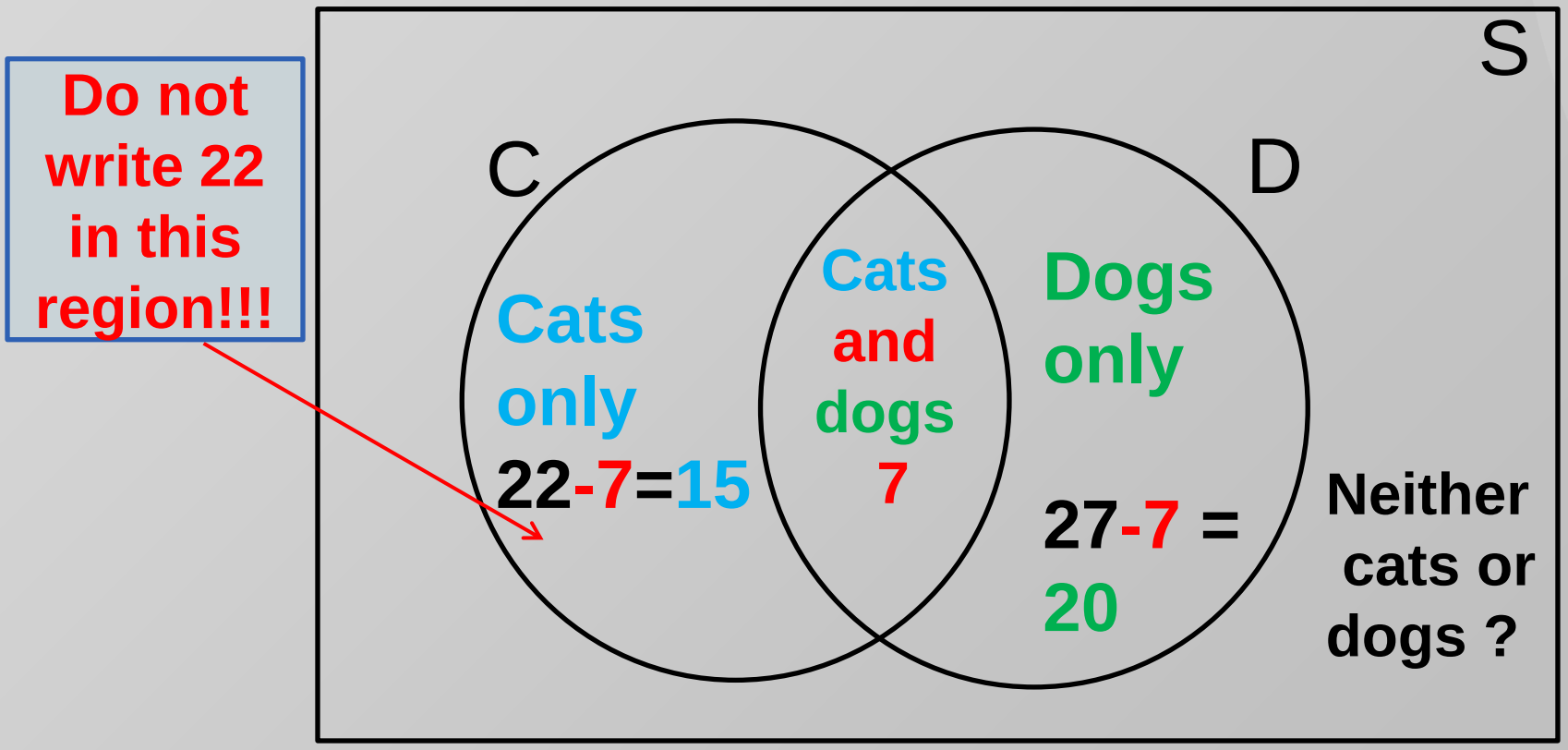
A family is chosen at random. Find the probability that chosen the family

b) owns dogs only,
c) does not own cats,
d) does not own dogs or cats.

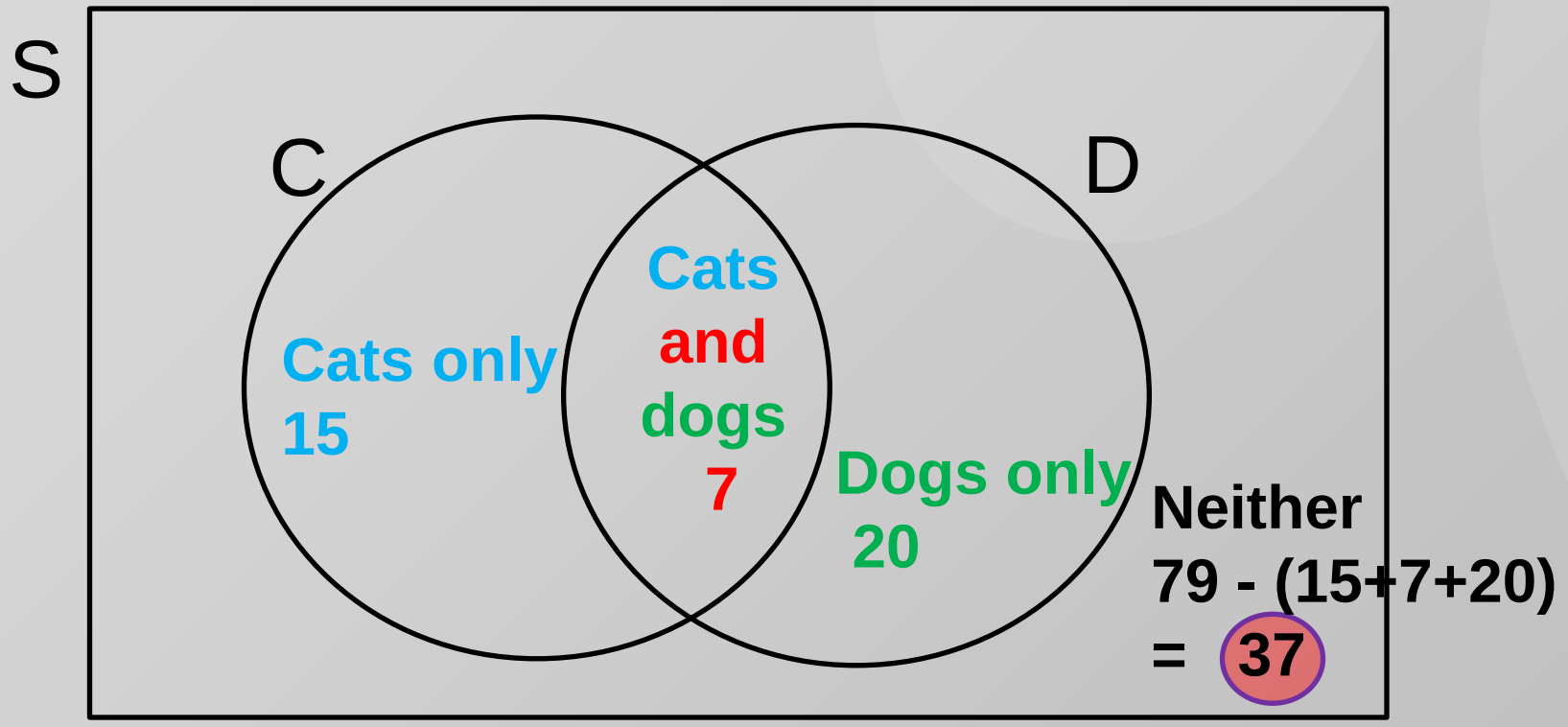
Solution

A survey of 79 families found that 22 own cats, 27 own dogs, and 7 own both cats and dogs.

$n(S) = 79$, $n(C) = 22$, $n(D) = 27$, $n(C \cap D) = 7$.



Solution continued



b) owns dogs only: $\frac{20}{79}$

c) does not own cats: $\frac{20+37}{79} = \frac{57}{79}$

d) does not own dogs or cats: $\frac{37}{79}$

Addition Rule (“or” probabilities)

Suppose a certain experiment is made. There is a sample space, S, and there are two events, A and B. What is the probability that **A or B** occurs?
It would make sense for this probability to be greater than both P(A) and P(B) alone.

The idea is that we want to add the number of outcomes in A with the number of outcomes in B, and then subtract off any that we have double-counted (i.e. those in $A \cap B$).

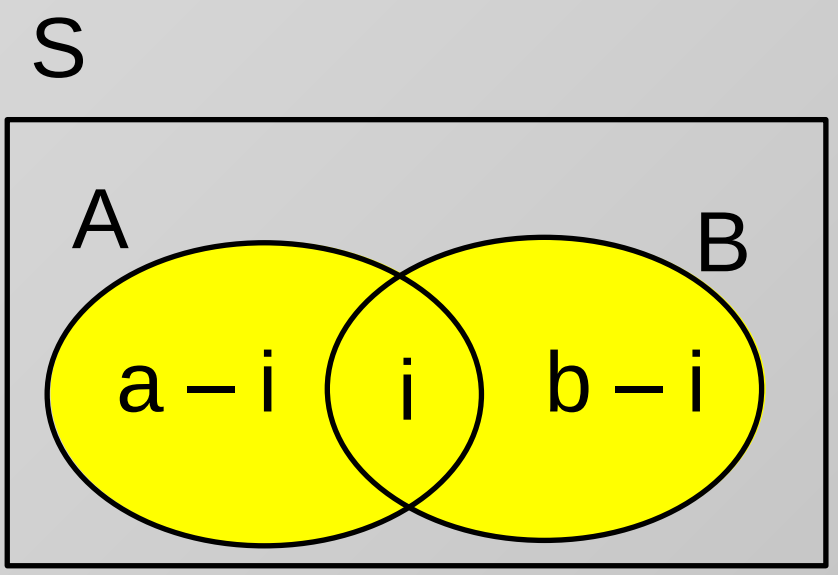
$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Addition Rule ("or" probabilities)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Consider a two event Venn diagram to see why this is true.

- Let the sample space be S and
- $P(A) = a$, $P(B) = b$, and $P(A \cap B) = i$



$P(A \cup B)$

$= a - i + i + b - i$

$= a + b - i$

$= P(A) + P(B) - P(A \cap B)$

Example 7

A and B are two events and $P(A) = 0.6$, $P(B) = 0.7$, and $P(A \cup B) = 0.9$.

Find

a) $P(A \cap B)$

b) $P(A')$

c) $P(A' \cup B)$

d) $P(A' \cap B)$

Solution

A and B are two events and P(A)=0.6, P(B)=0.7 and P(A ∪ B)=0.9 .

Find

a) $P(A \cap B) = P(A) + P(B) - P(A \cup B)$

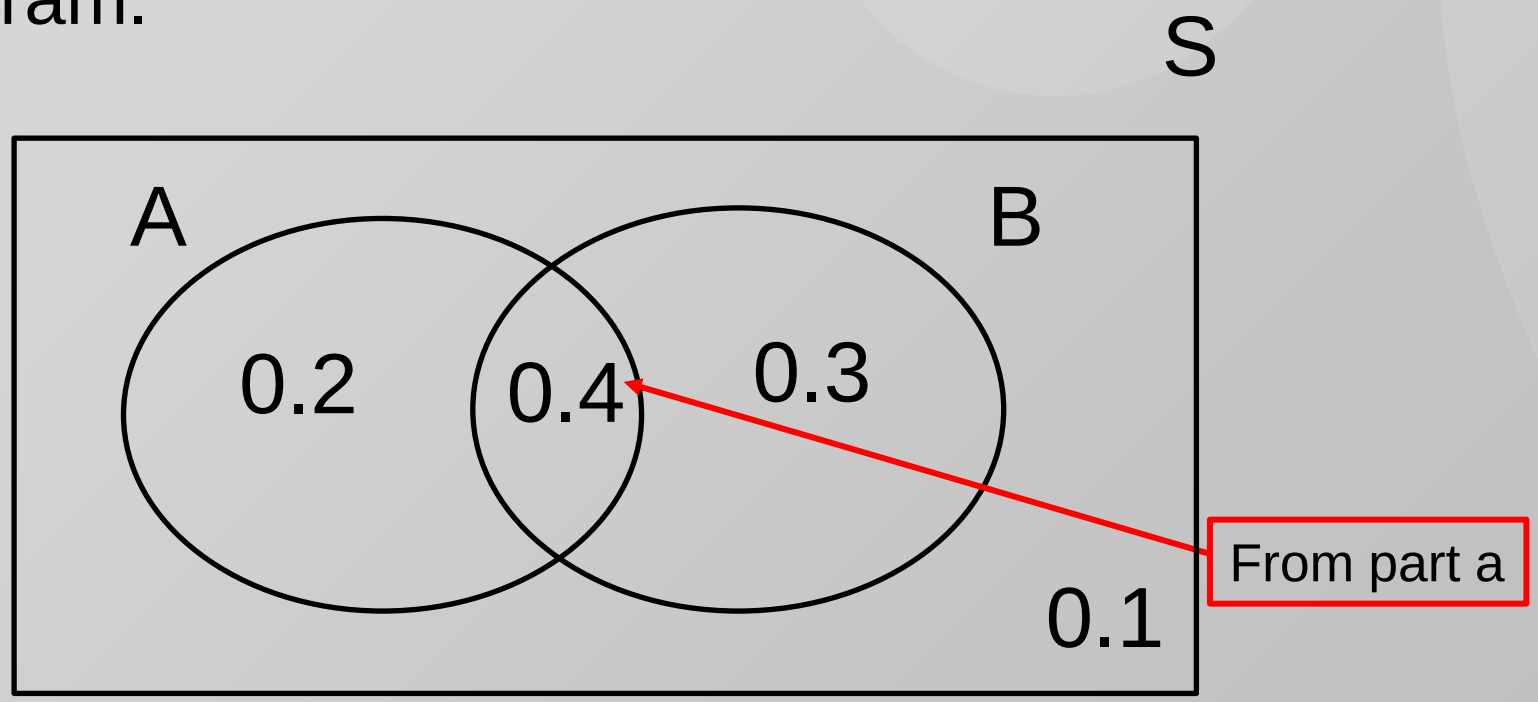
$= 0.6 + 0.7 - 0.9 = \mathbf{0.4}$

b) $P(A') = 1 - P(A)$

$= 1 - 0.6 = \mathbf{0.4}$

Solution (continued)

To answer parts c) and d), we will use a Venn diagram.



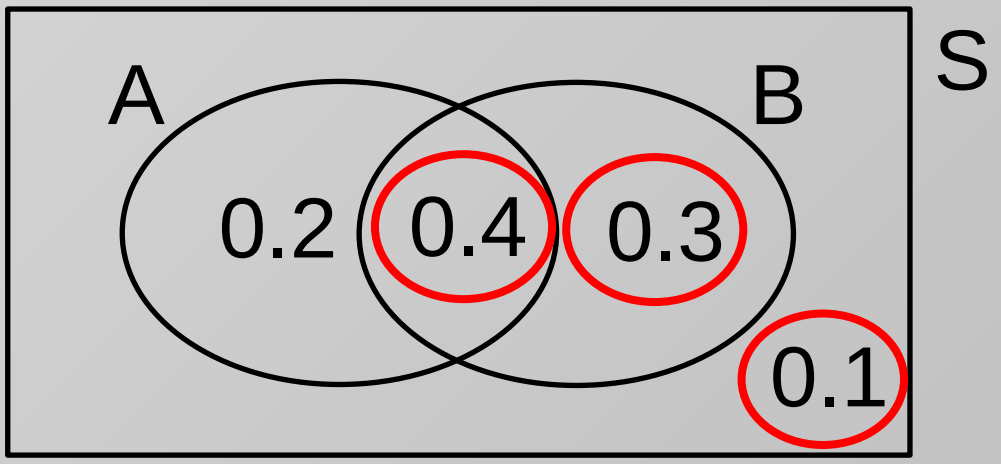
Given: P(A) = 0.6, P(B) = 0.7, and P(S) = 1

Solution (continued)

c) P(A’ U B)

This means we want to be *not* in A ***or*** in B.

Find this area on the Venn diagram:



So, P(A’ U B) = 0.4 + 0.3 + 0.1 = **0.8**

Solution (continued)

d) $P(A' \cap B)$

<u>Method 1</u> Using the formula:

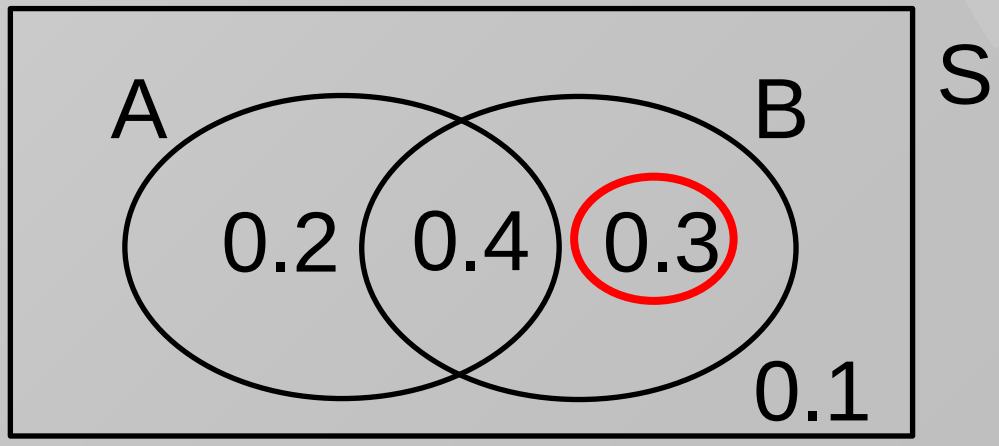
$P(A' \cap B) = P(A') + P(B) - P(A' \cup B)$

$= 1 - 0.6 + 0.7 - 0.8 = \mathbf{0.3}$

<u>Method 2</u> We want to be *not* in A ***and*** in B.

Find this area on the Venn diagram:

So, $P(A' \cap B) = \mathbf{0.3}$



Your turn!

A and B are two events and $P(A)=0.4$, $P(B)=0.5$ and $P(A \cup B)=0.6$.

Find

a) $P(A \cap B)$

b) $P(A')$

c) $P(A \cup B')$

d) $P(A' \cup B)$

Solution

a) $P(A \cap B) = P(A) + P(B) - P(A \cup B)$

$= 0.4 + 0.5 - 0.6 = \mathbf{0.3}$

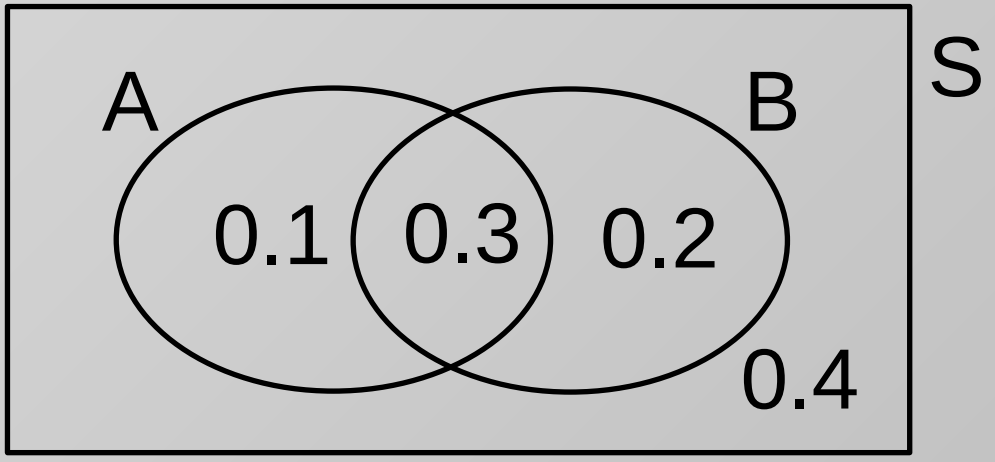
b) $P(A') = 1 - P(A) = 1 - 0.4 = \mathbf{0.6}$

Solution continued

c) P(A U B’)

This means we want to be in A ***or*** not in B.

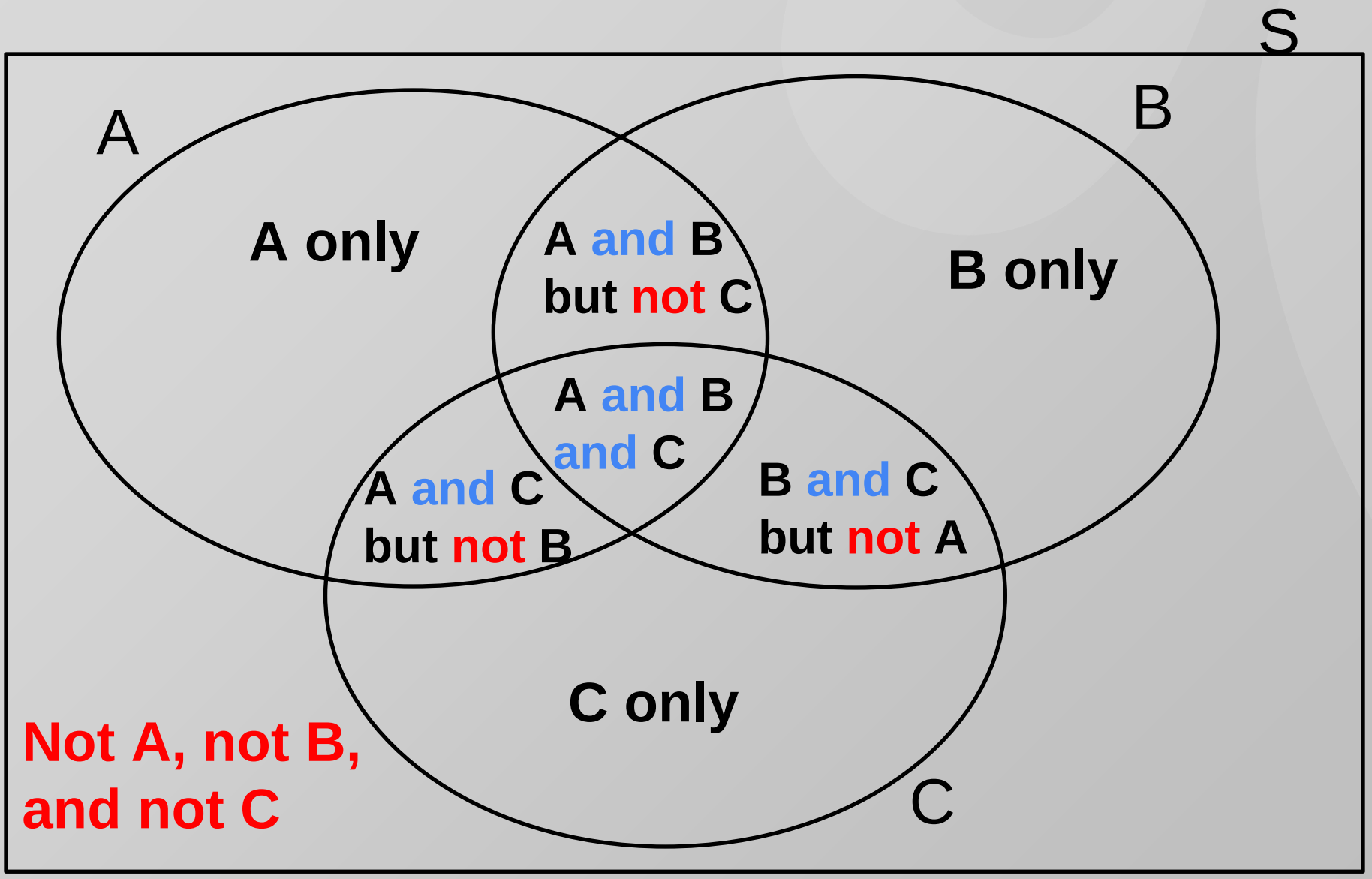
Find this area on the Venn diagram:



So, P(A U B’) = 0.1 + 0.3 + 0.4 = **0.8**

d) Similarly, P(A’ U B) = 0.2 + 0.4 + 0.3 = **0.9**

Venn diagram with three events



Example 8

A vet surveys 100 of her clients. She finds that 25 own dogs; 15 own dogs and cats; 11 own dogs and tropical fish; 53 own cats; 10 own cats and tropical fish; 7 own dogs and cats and tropical fish; 40 own tropical fish.

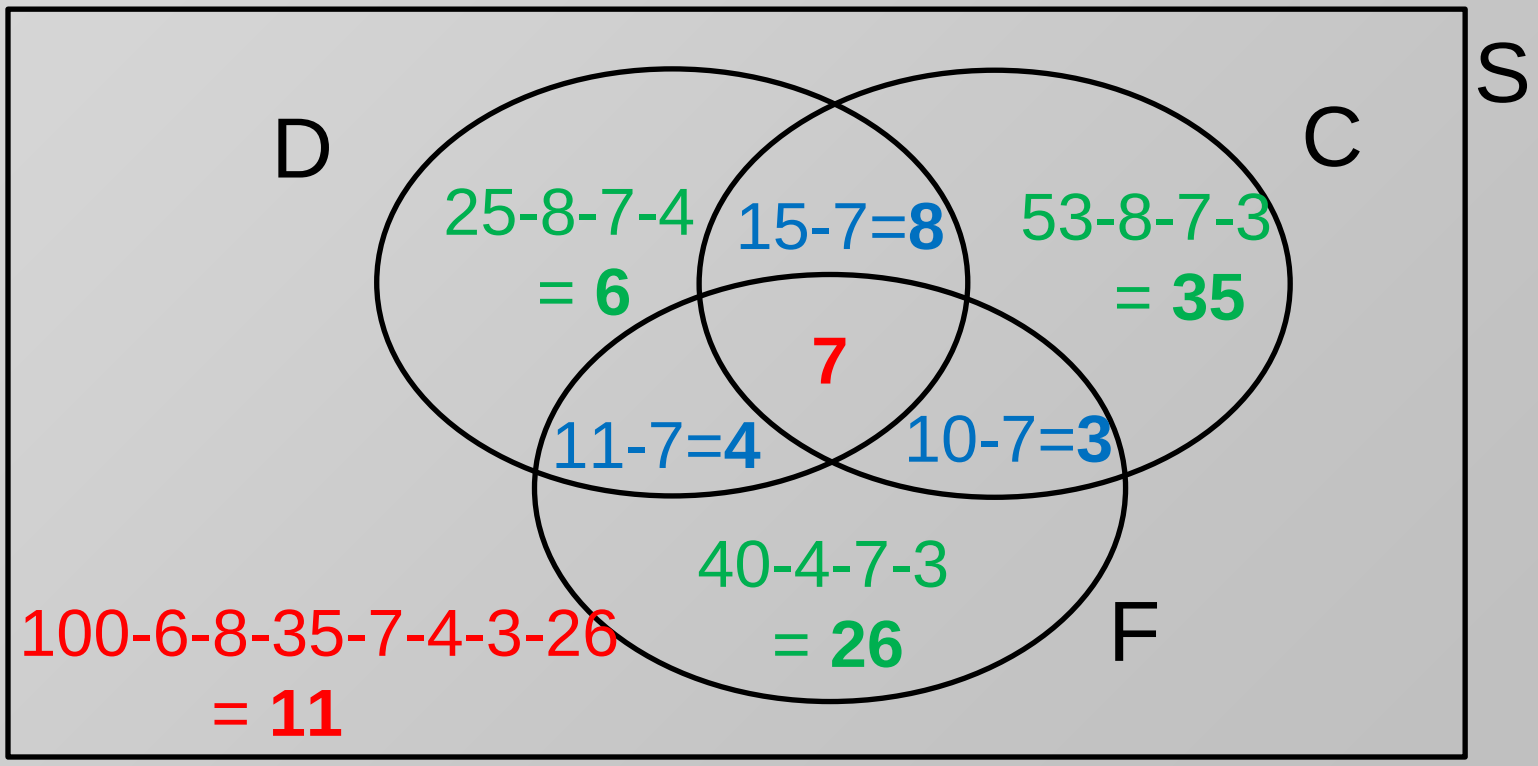
a) Draw a Venn diagram to represent this information.

A client is chosen at random. Find the probability that the client

b) owns dogs only,
c) does not own tropical fish,
d) does not own dogs, cats, or tropical fish.

Solution

A vet surveys 100 of her clients. She finds that 25 own dogs, 15 own dogs and cats, 11 own dogs and tropical fish, 53 own cats, 10 own cats and tropical fish, 7 own dogs, cats and tropical fish, 40 own tropical fish.

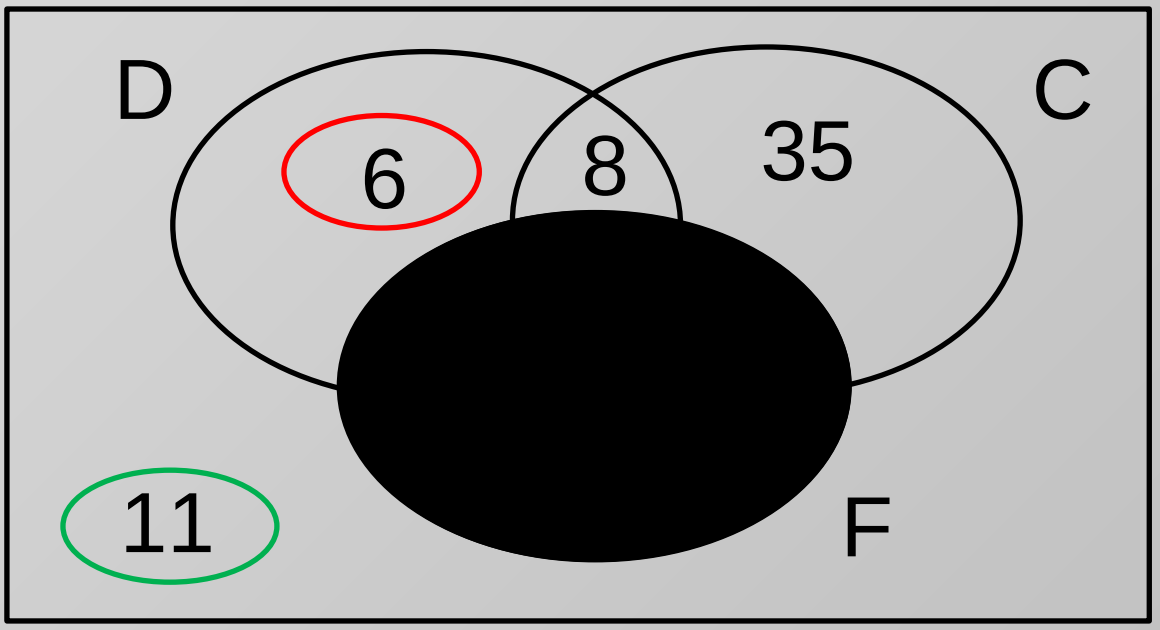


Solution (continued)

b) owns dogs only $\frac{6}{100} = \mathbf{0.06}$

c) does not own tropical fish, $\frac{11+6+8+35}{100} = \mathbf{0.6}$

d) does not own dogs, cats, or tropical fish. $\frac{11}{100} = \mathbf{0.11}$

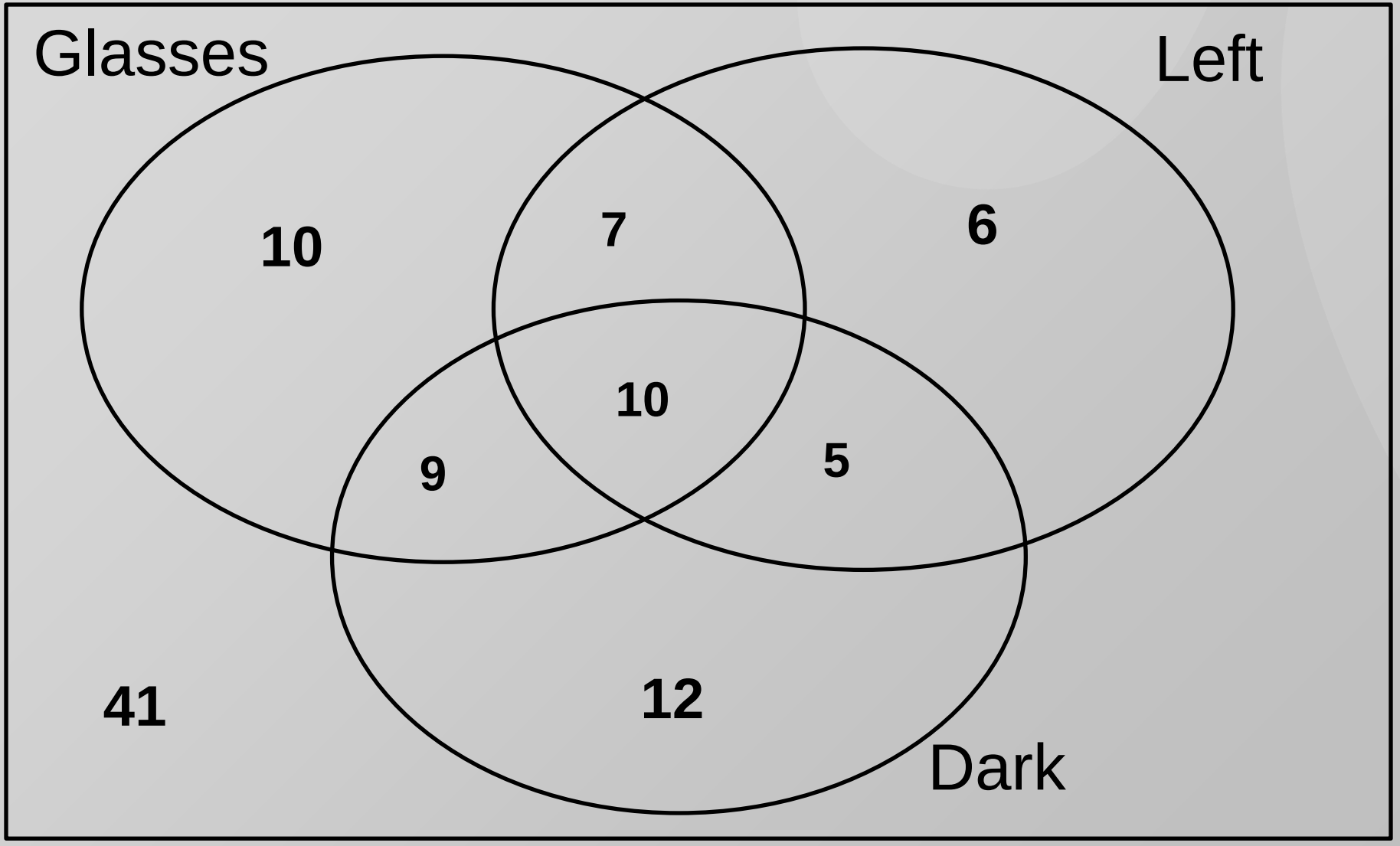


Your turn!

A group of 100 people produced the following information relating to three attributes.
The attributes are wearing glasses, being left-handed and having dark hair.
Glasses are worn by 36 people, 28 people are left-handed, and 36 people have dark hair.
There are 19 people who wear glasses and have dark hair, 17 people who wear glasses and are left-handed, and 15 people who are left-handed and have dark hair.
Only 10 people wear glasses, are left-handed, and have dark hair.

a) Represent these data on a Venn diagram.

Solution



Your turn!

Based on the Venn diagram you drew, solve the following questions.

A person was selected at random from this group. Find the probability that this person

b) Wears glasses but is not left-handed and does not have dark hair

c) Does not wear glasses, is not left-handed and does not have dark hair

d) Has only two attributes

Solution

b) $\frac{10}{100} = \mathbf{0.1}$

c) $\frac{41}{100} = \mathbf{0.41}$

d) $\frac{7+9+5}{100} = \frac{21}{100} = \mathbf{0.21}$

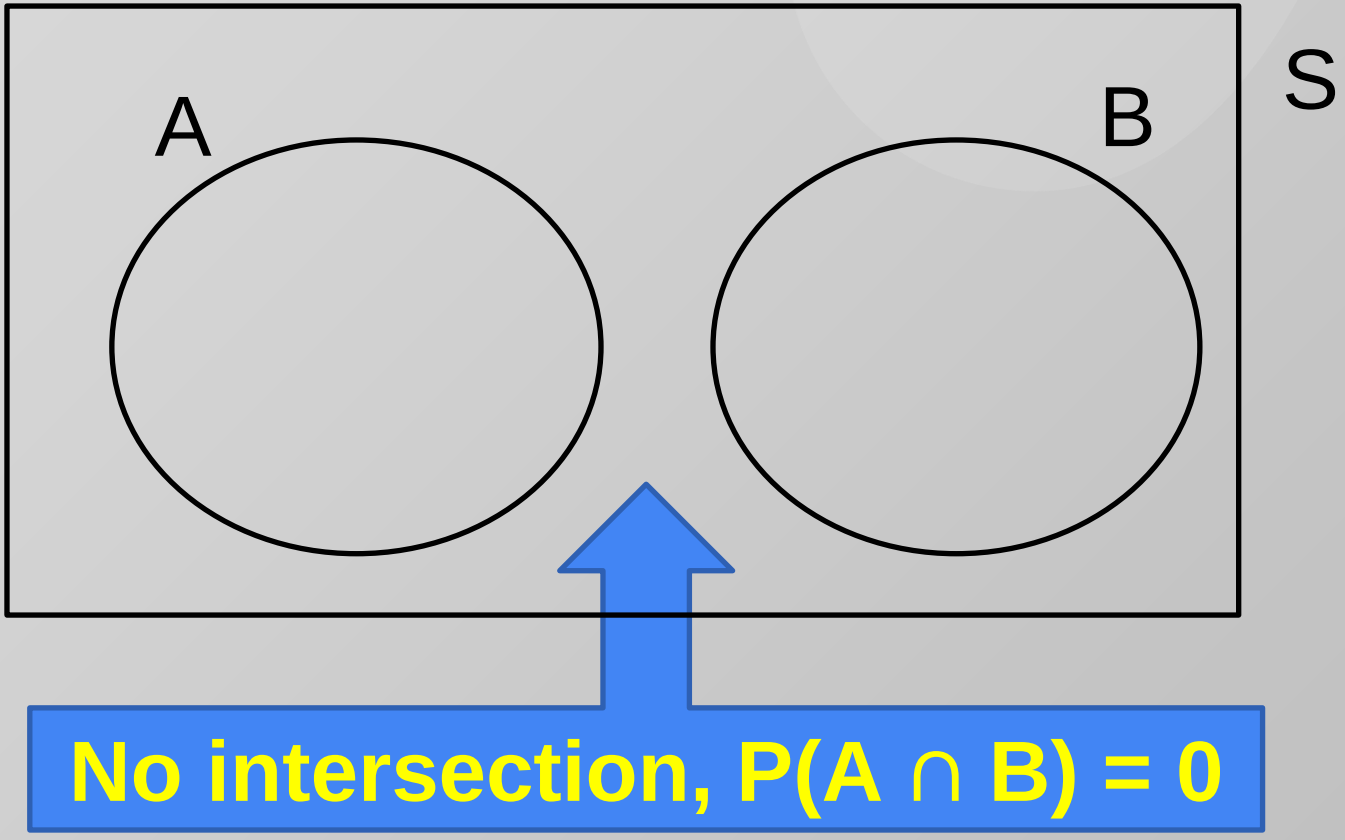
Mutually exclusive events

- When **events have no outcomes in common**, then they are **mutually exclusive**.

- This means that the events cannot occur together.
- e.g.

1) The event it is raining and the event it is not raining (at the same time, in the same place)
2) The event of getting an even number and the event of getting an odd number

Mutually exclusive events in Venn diagram



Can you state the addition rule for mutually exclusive events?

Addition rule for mutually exclusive events

$$P(A \cup B) = P(A) + P(B) - \cancel{P(A \cap B)}$$

0

If events A and B are mutually exclusive, then

$$P(A \cup B) = P(A) + P(B)$$

Example 9

Events A and B are mutually exclusive and $P(A) = 0.2$ and $P(B) = 0.4$.

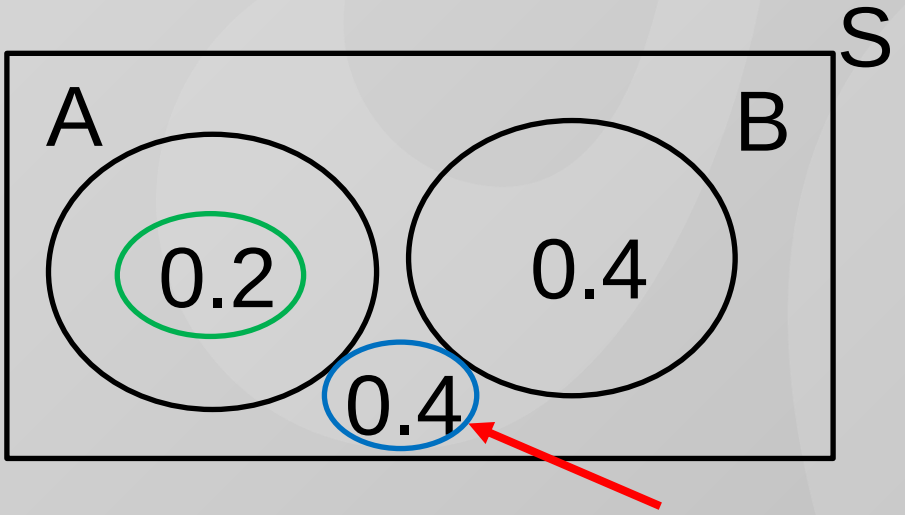
Find

a) $P(A \cup B)$

b) $P(A \cap B')$

c) $P(A' \cap B')$

Solution



$1 - 0.2 - 0.4 = 0.4$

a) P(AUB)

= P(A) + P(B)

= 0.2 + 0.4 = **0.6**

b) P(A ∩ B')

*in A **and** not in B* :

P(A ∩ B') = **0.2**

c) P(A' ∩ B')

*not in A **and** not in B* :

P(A' ∩ B') = **0.4**

Formulae

- $P(A') = 1 - P(A)$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- $P(A \cup B) = P(A) + P(B)$

if A and B are mutually exclusive

Learning outcomes

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Preview: Probability 2

Independent events

- Multiplication Rule

Dependent Events (conditional probability)

- Using the Multiplication Rule
- Using a tree diagram
- Using a Venn diagram